

3. (a) Most of the tap water in Wales comes from mountain reservoirs. The water from these reservoirs is never pure.

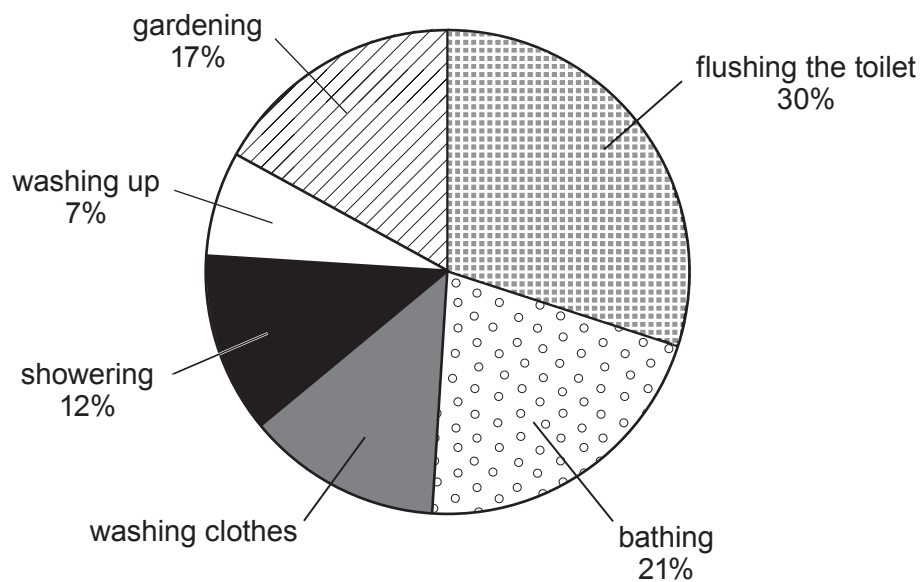
- (i) Information about some of the stages used in the treatment of tap water is shown.

Draw a line to connect each stage to the correct description. One has already been completed. [2]

Stage	Description
filtration	large insoluble particles are allowed to settle out as sludge
screening	screens are used to remove objects such as leaves, sticks and litter
chlorination	harmful bacteria are killed to make the water safe to drink
sedimentation	small insoluble particles are removed by passing the water through sand beds



- (ii) The pie chart shows some of the main uses of water in a typical Welsh home.



Use the pie chart to answer parts I and II.

- I. Calculate the percentage of water used for washing clothes. [1]

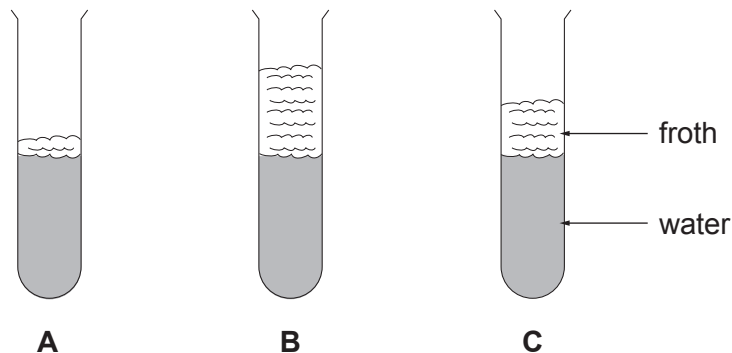
..... %

- II. Give the name of the use that accounts for three times as much water as another use. [1]

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- (b) 5 cm³ of three water samples, labelled **A**, **B** and **C**, were mixed with 10 drops of soap solution and shaken. One of the samples was distilled water. The boiling point of each water sample was also measured. The results obtained are shown.



Water sample	Boiling point (°C)
A	102
B	100
C	101

- (i) Place the water samples, **A**, **B** and **C**, in order of hardness.

[1]

hardest

.....

softest

- (ii) Give the letter of the sample that is distilled water. Give **two** reasons for your choice.

[2]

Sample

Reasons

1.

.....

2.

.....



- (iii) The volume of each water sample and soap solution was kept the same for each test. State which other variable must also be controlled before measuring the froth height. Describe how you would control this. [2]

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- (iv) Hardness in water can be caused by dissolved calcium fluoride. Give the chemical formula for calcium fluoride. [1]

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